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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025
ME1104-1 – THERMAL ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A closed system**
 - A) is a specified region where transfer of energy and /or mass take place
 - B) is a region of constant mass and only energy is allowed to cross the boundaries
 - C) cannot transfer either energy or mass to or from the surroundings
 - D) is one in which mass within the system is not necessarily constant
2. **Which of the following is not a property of the system?**
 - A) Specific volume
 - B) Temperature
 - C) Pressure
 - D) Heat
3. **Internal energy of a perfect gas depends on**
 - A) temperature, specific heats, and pressure
 - B) temperature, specific heats, and enthalpy
 - C) temperature, specific heats, and entropy
 - D) temperature only.
4. **A throttling process is always an**
 - A) Isochoric process
 - B) Isenthalpic process
 - C) Isothermal process
 - D) Isentropic process
5. **In a nozzle, approach velocity is**
 - A) lesser than exit velocity
 - B) greater than exit velocity
 - C) equal to exit velocity
 - D) may be lesser or greater than exit velocity
6. **The characteristic gas constant (R) is equal to the**
 - A) sum of two specific heats
 - B) difference of two specific heats
 - C) product of two specific heats
 - D) ratio of two specific heats
7. **First law of thermodynamics deals with**
 - A) temperature
 - B) energy
 - C) enthalpy
 - D) entropy
8. **The latent heat of vaporization at critical point is**
 - A) less than zero
 - B) greater than zero
 - C) equal to zero
 - D) doesn't have a definite value
9. **Choose the correct statement**
 - A) Critical point involves equilibrium of solid and vapor phases
 - B) Critical point involves equilibrium of solid and liquid phases
 - C) Triple point involves equilibrium of solid, liquid and vapour phases.
 - D) Critical point involves equilibrium of solid, liquid and vapour phases
10. **The specific volume of water when heated at 0° C**
 - A) first decreases and then increases
 - B) first increases and then decreases
 - C) decreases steadily
 - D) increases steadily
11. **Rankine cycle comprises of**
 - A) two isobaric and two isochoric processes
 - B) two isothermal and two isobaric processes
 - C) two isobaric and two isentropic processes
 - D) two isochoric and two isentropic processes

12. In Rankine cycle the work input to the pump is given by
 A) change of internal energy between outlet and inlet
 B) change of temperature between outlet and inlet
 C) change of entropy between outlet and inlet
 D) change of enthalpy between outlet and inlet
13. For the same compression ratio
 A) thermal efficiency of Diesel cycle is greater than that of Otto cycle
 B) thermal efficiency of Diesel cycle is less than that of Otto cycle
 C) thermal efficiency of Diesel cycle is same as that of Otto cycle
 D) Thermal efficiency of Diesel cycle cannot be predicted
14. In air standard diesel cycle, at fixed compression ratio and fixed value of adiabatic index (γ)
 A) thermal efficiency increases with increase in fuel cut-off ratio
 B) thermal efficiency decreases with increase in fuel cut-off ratio
 C) thermal efficiency remains same with increase in fuel cut-off ratio
 D) may increase or decrease with increase in fuel cut-off ratio
15. The air standard efficiency of Brayton cycle is given by (r_p – pressure ratio)
 A) $r_p^{\frac{\gamma-1}{\gamma}} - 1$
 B) $1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$
 C) $1 - r_p^{\gamma-1}$
 D) $1 - \frac{1}{r_p^{\gamma-1}}$
16. Thermal efficiency of closed cycle gas turbine power plant increases by
 A) Reheating
 B) Regeneration
 C) Intercooling
 D) Reheating, Regeneration or Intercooling
17. Out of all the refrigeration systems, the one with most importance from the standpoint of commercial and domestic utility is
 A) Open air refrigeration system
 B) Closed air refrigeration system
 C) Vapor compression system
 D) Vapor absorption system
18. The function of providing heat transfer surface through which heat transfers from hot refrigerant vapor to the cooling medium is done by
 A) compressor
 B) condenser
 C) throttle valve
 D) evaporator
19. Degree of saturation is defined as the ratio of
 A) actual specific humidity to the saturated specific humidity
 B) mass of dry air to mass of water vapor in the mixture
 C) mass of water vapor to mass of dry air in the mixture
 D) actual mass of vapor to the mass of saturated vapor
20. The relative humidity during heating and dehumidification
 A) increases
 B) decreases
 C) remains same
 D) may increase or decrease

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Differentiate the following with suitable examples: i. Intensive property and Extensive property ii. Homogeneous system and Heterogeneous system iii. Point function and Path function iv. Thermodynamic process and Thermodynamic cycle	8	L*1	1	1
	b) Establish the relation between Fahrenheit scale and Rankine scale using two fixed point method	8	L2	2	1
2.	a) Derive the expressions for heat transfer in i. Isobaric process ii. Polytropic process	8	L2	1	1

- b) A system undergoes various processes in which the initial state is identical to the final state. The work and heat interactions during the processes are tabulated below. Complete the table and calculate net work done by the system.

Process	Q (kJ/s)	W (kJ/s)	ΔU (kJ/s)
1 - 2	---	35.13	64.85
2 - 3	50	25	---
3 - 4	- 50	---	---
4 - 1	---	17.6	- 17.57

3. a) Define the terms COP of Refrigerator and that of Heat pump. Also establish the relation between them.
- b) Two Carnot heat engines A and B are connected in series. Engine A is rejecting heat to engine B through an intermediate reservoir. Engine A receives 500 kJ from a reservoir at 450°C while engine B is in communication with a sink at a temperature at 10°C. If the efficiency of engine A is twice that of engine B. Estimate
- Intermediate temperature between A and B
 - Work output of engines A and B.
 - Heat rejected to sink.

4. a) Draw the P-T diagram for water. Indicate all the salient points/regimes.
- b) The pressure and temperature of the air in the diesel cycle are 1 bar and 27°C respectively before the compression stroke. If the compression ratio is 18 and the cycle peak temperature is 2000°C, obtain the air standard efficiency of the cycle. Also find out the pressure and temperature at all the salient points.
5. a) A Rankine cycle using water as the working fluid operates between the pressure limits of 10 kPa and 200 bar. The maximum temperature of the cycle is 700°C. Determine cycle thermal efficiency and the steam rate.
- b) Derive the expression for optimum pressure ratio and corresponding maximum efficiency in Brayton cycle.
6. a) Explain the processes in an Ideal Reheat Rankine cycle with schematic and T-S diagrams.
- b) Derive the expression for air standard efficiency of Otto cycle.

Unit - III

7. a) A 10-ton ammonia ice plant operates between an evaporator temperature of -20°C and a condenser temperature of 30°C. The ammonia enters the compressor as dry saturated vapor. Assuming isentropic compression, determine;
- mass flow rate of NH_3
 - COP of the plant
 - power input in kW and
 - tons of ice at -10°C produced from water at 25°C in a day.
- [Enthalpy of fusion of ice = 335 kJ/kg, C_p of water = 4.187 kJ/kg K, C_p of ice = 2.1 kJ/kg K and C_p of superheated Ammonia vapor = 2.76 kJ/kg K]

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|-------|--|---|----|---|---|
| b) | Explain Bell-Coleman cycle with schematic and T – S diagrams, | 6 | L1 | 5 | 1 |
| 8. a) | Define the terms humidity ratio and Relative humidity. Also derive the expression relating them. | 8 | L2 | 5 | 1 |
| b) | Atmospheric air at 101.325 kPa has 30°C DBT and 15°C DPT. Without using the psychrometric chart, using the property values from the tables, calculate, | 8 | L3 | 5 | 2 |
| | i. Partial pressures of air and water vapor | | | | |
| | ii. Specific humidity | | | | |
| | iii. Relative humidity | | | | |
| | iv. Vapor density | | | | |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
